

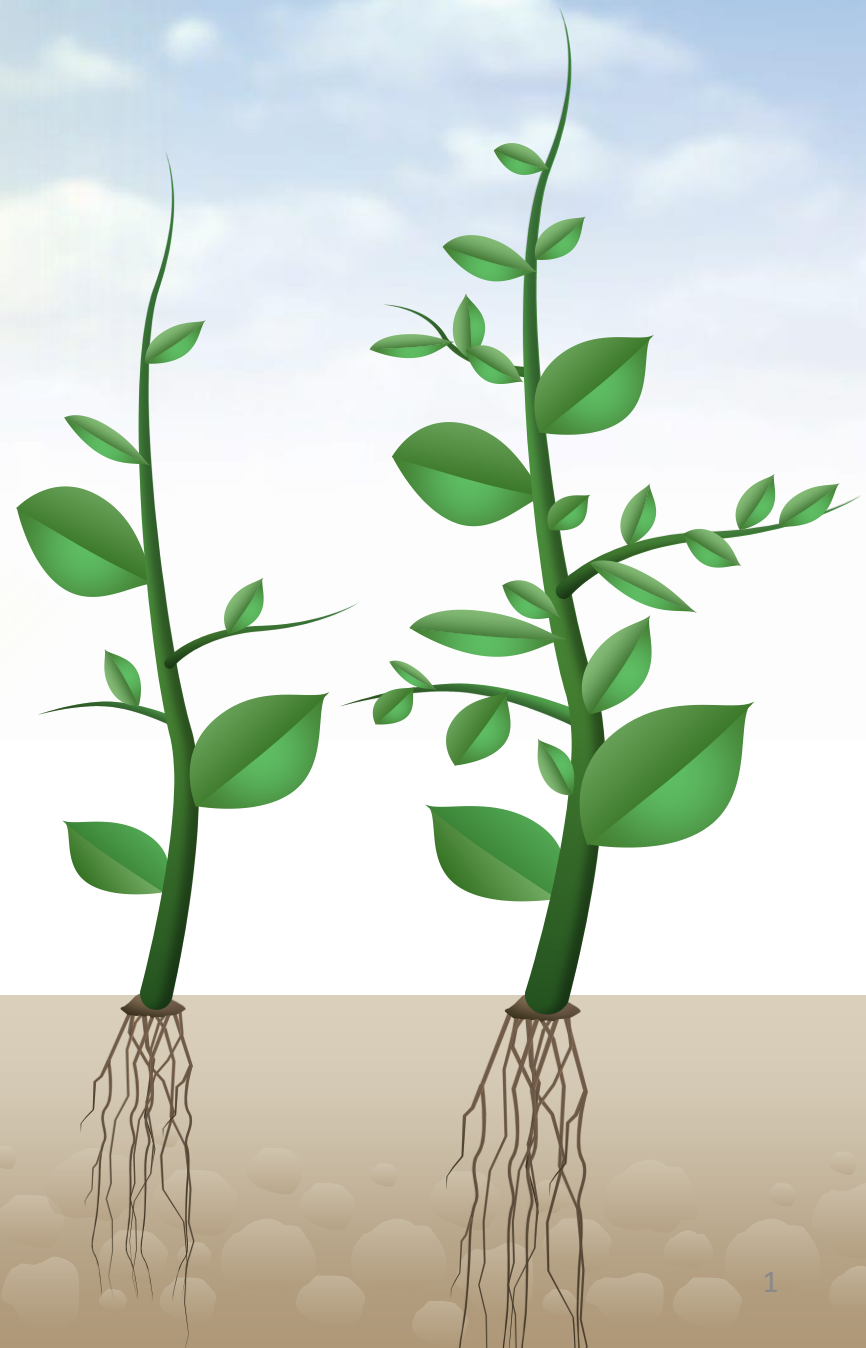
Dynamics of Particles

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Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

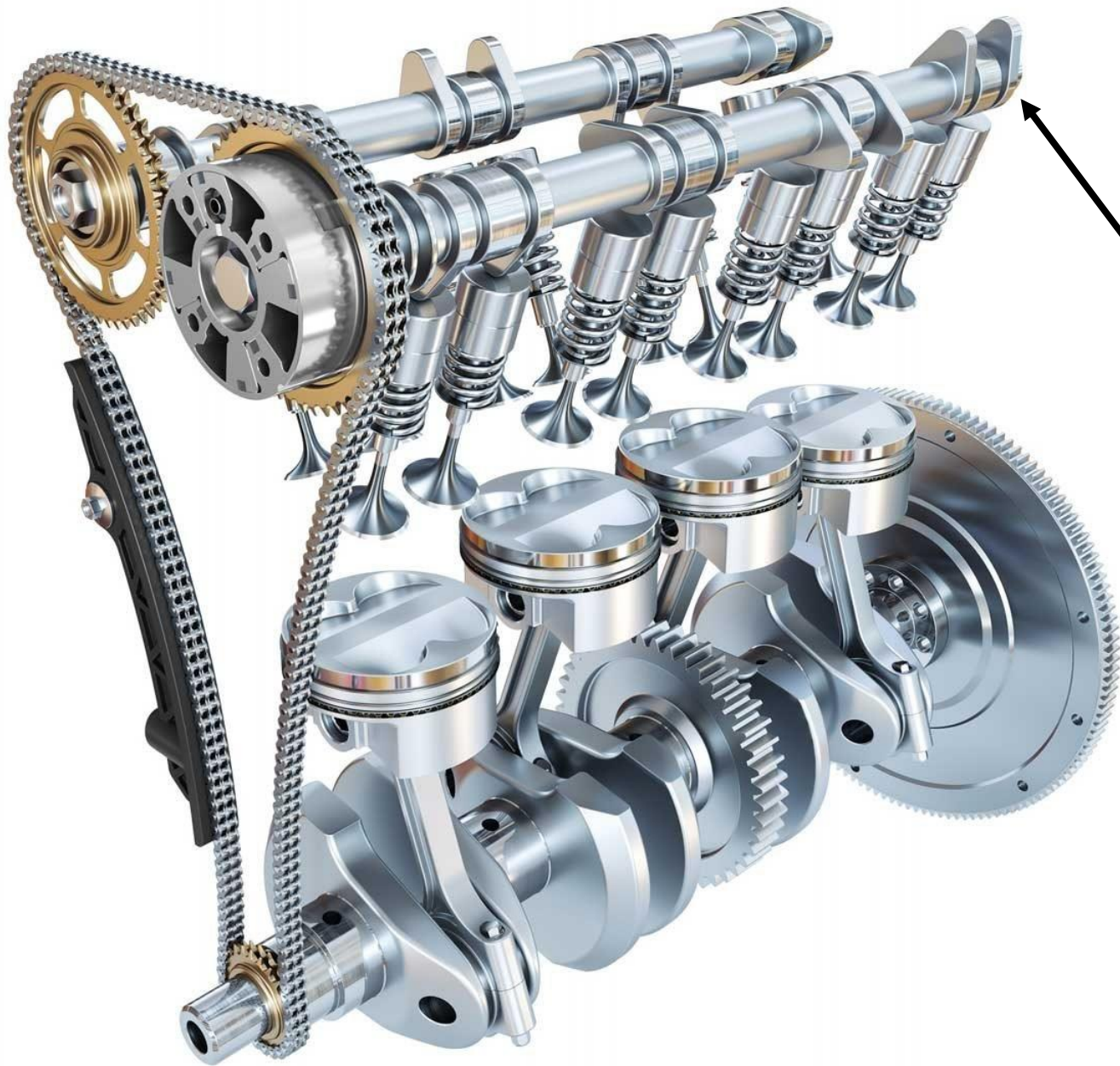
Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Kinematics of Particles

■ What is Kinematics?

- Kinematics is the **branch of dynamics** which describes the **motion of bodies without reference to the forces** which either cause the motion or are generated as a result of the motion
- **Geometry of Motion**
- **Engineering applications of Kinematics**
 - Design of cams, gears, linkages, and other machine elements to control or produce certain desired motions
 - Calculation of flight trajectories for aircraft, rockets, and spacecraft
- **Prerequisite for kinetics**



Cam Shaft

Kinematics of Particles

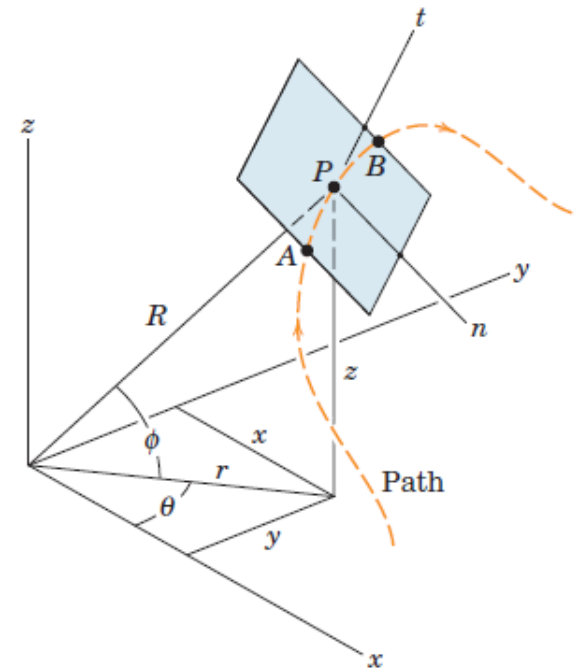
■ Particle?

- A particle is a body whose physical dimensions are so small compared with the radius of curvature of its path that we may treat the motion of the particle as that of a point
- **Example:** Wingspan of a jet transport flying between Jodhpur and Kerala

Coordinate Systems

■ Coordinate systems

- Rectangular Coordinate (x-y-z)
- Cylindrical Coordinate (r- θ -z)
- Spherical Coordinate (R- θ - ϕ)
- Normal-tangential coordinates:
 - measurements along the tangent t and normal n to the curve
 - The direction of n lies in the **local plane of the curve**
 - These measurements are called **PATH VARIABLES**.

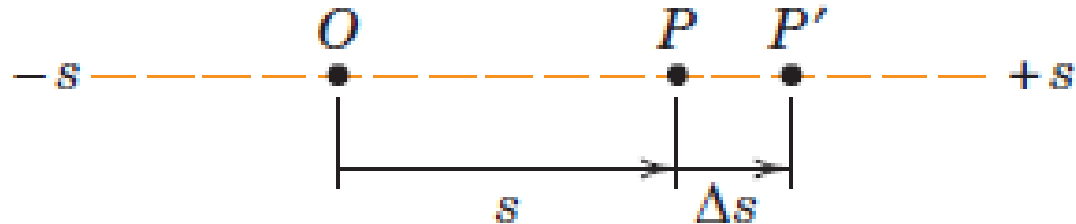


Plane motion

- **Plane motion**
 - All movements occur in or can be represented as occurring in a single plane
 - In many cases, motions of machines and structures in engineering can be represented as plane motion

Rectilinear motion

■ Motion along a line



- Average velocity

$$v_{av} = \frac{\Delta s}{\Delta t}$$

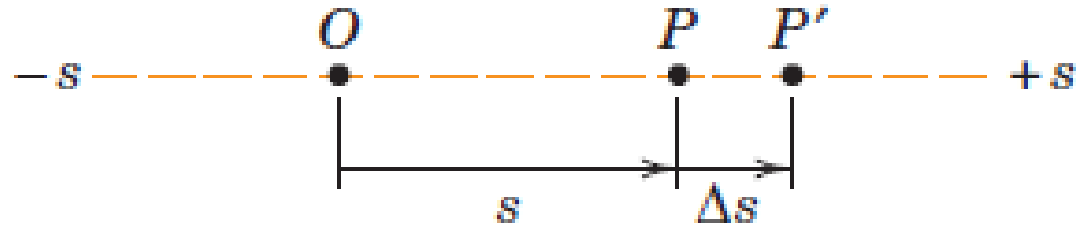
- Instantaneous velocity

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$$

$$v = \frac{ds}{dt}$$

Time rate of change of position coordinate, s

Rectilinear motion



- Average acceleration

$$a_{av} = \frac{\Delta v}{\Delta t}$$

- Instantaneous acceleration

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

$$a = \frac{dv}{dt}$$

$$a = \frac{d^2 s}{dt^2}$$

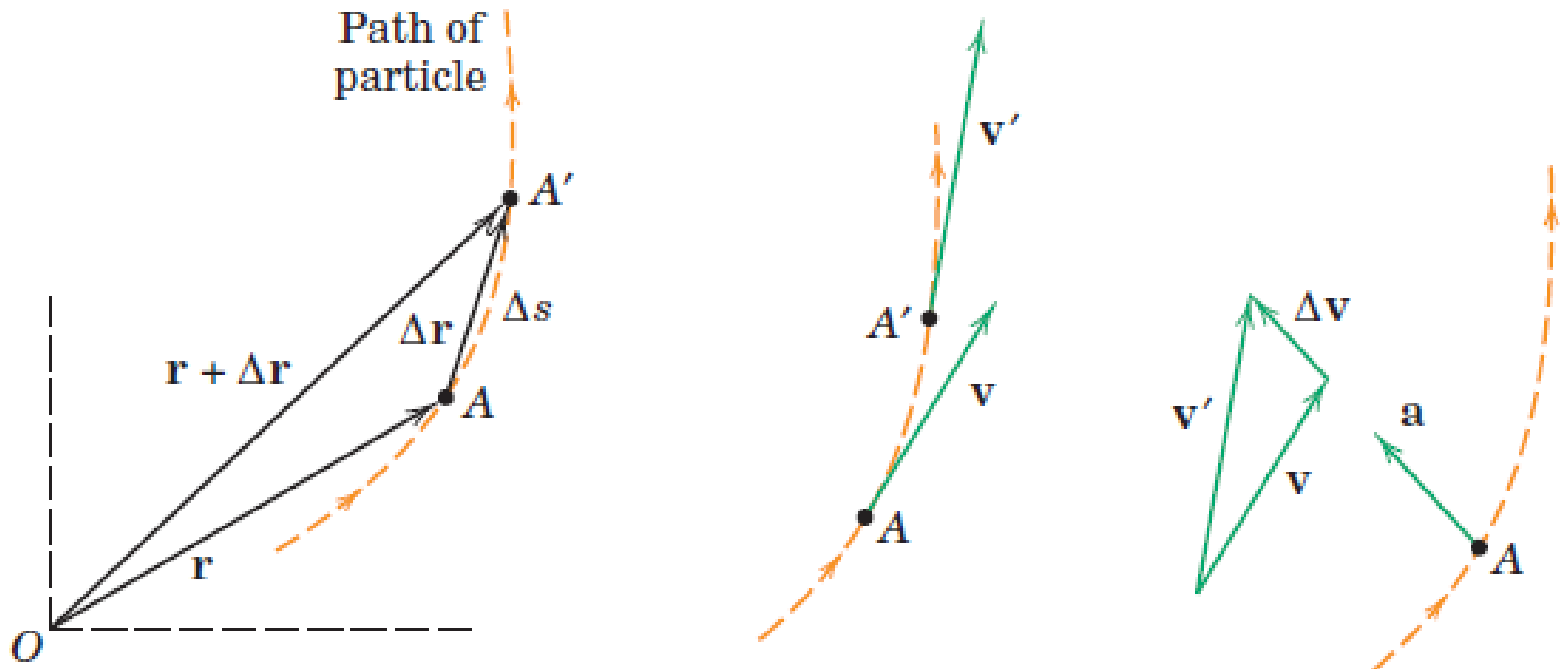
Time rate of change of velocity, v

$$v dv = a ds$$

$$\dot{s} d\dot{s} = \ddot{s} ds$$

Plane Curvilinear motion

- Motion along a plane curve



- Average velocity $\mathbf{v}_{av} = \frac{\Delta \mathbf{r}}{\Delta t}$

Plane Curvilinear motion

■ Motion along a plane curve

- Instantaneous velocity

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t}$$

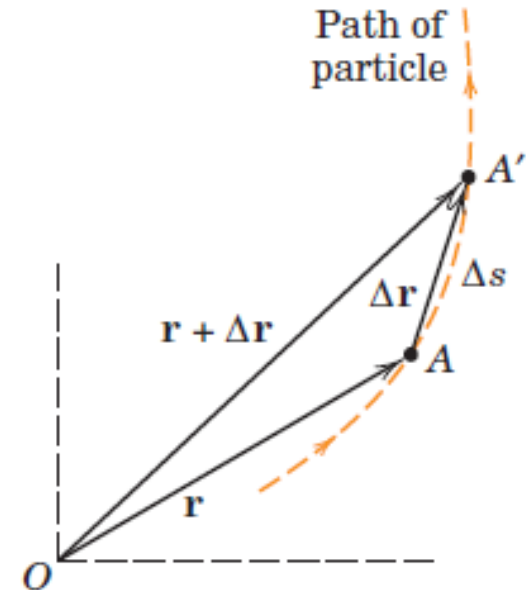
$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{\mathbf{r}}$$

- Average speed

$$\text{speed}_{av} = \frac{\Delta s}{\Delta t}$$

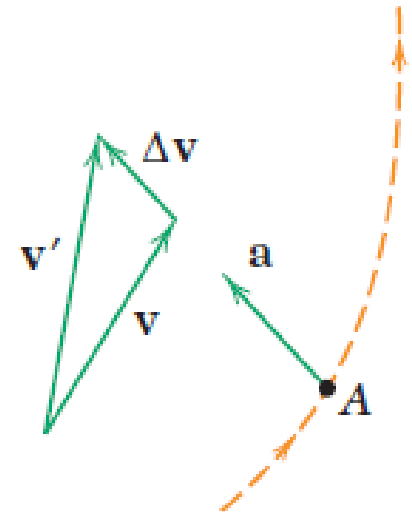
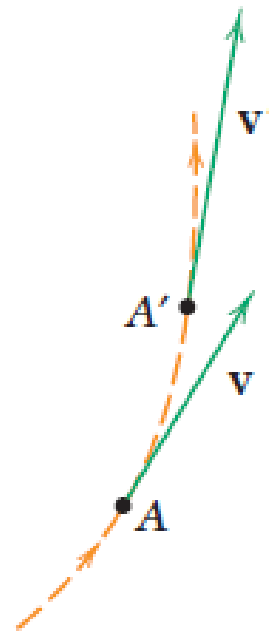
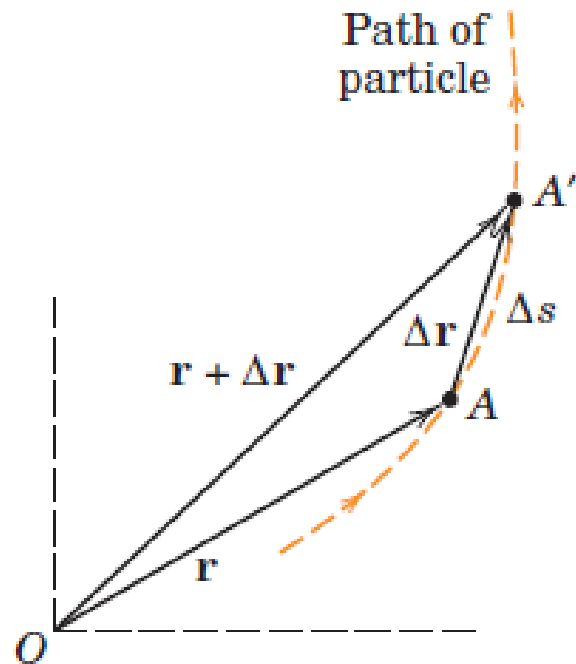
- Speed

$$v = |\mathbf{v}| = \frac{ds}{dt} = \dot{s}$$



Plane Curvilinear motion

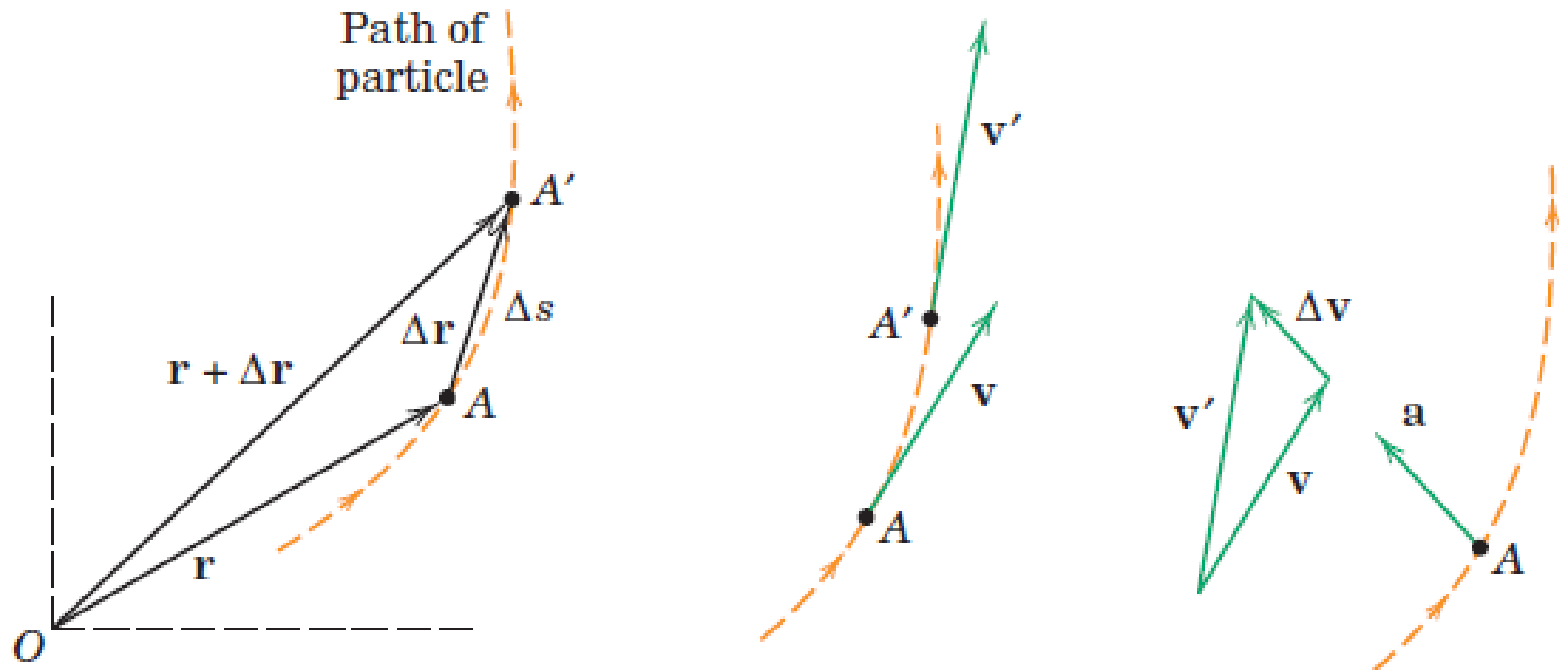
- Motion along a plane curve



- Average acceleration

$$\mathbf{a}_{av} = \frac{\Delta \mathbf{v}}{\Delta t}$$

Plane Curvilinear motion



- Instantaneous acceleration

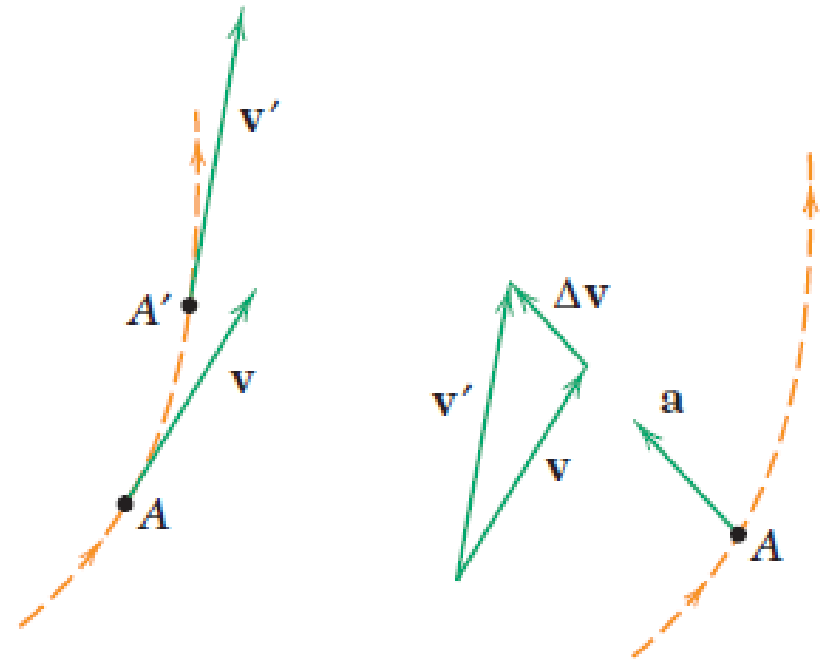
$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t}$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \dot{\mathbf{v}}$$

Plane Curvilinear motion

■ Motion along a plane curve

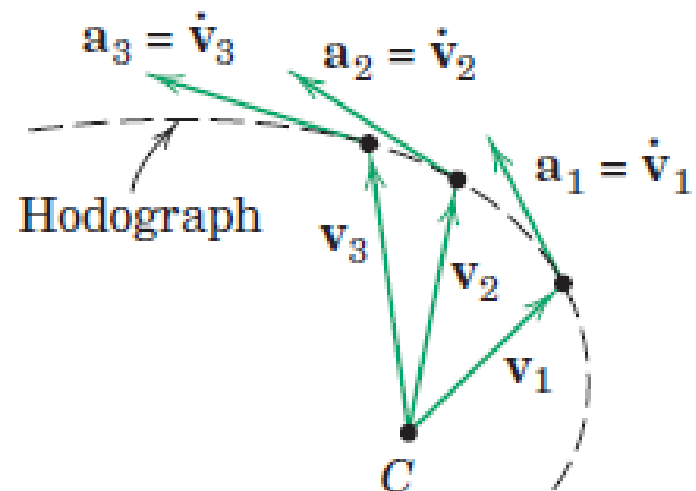
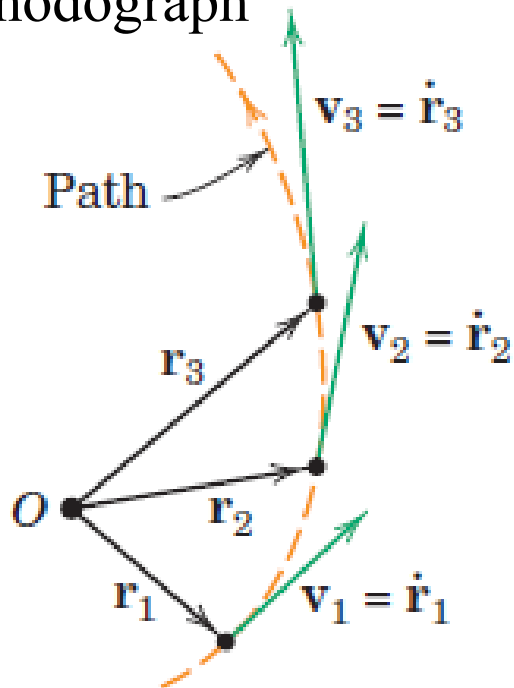
- Average acceleration
- Instantaneous acceleration
 - In general, the direction of the acceleration of a particle in curvilinear motion is neither tangent to the path nor normal to the path.
 - Acceleration *component which is normal to the path points toward the center of curvature of the path.*



Plane Curvilinear motion

■ Hodograph

- Velocity vectors plotted from some arbitrary point C , a curve, called the hodograph, is formed. The derivatives of these velocity vectors will be the acceleration vectors which are tangent to the hodograph



Rectangular Coordinates

■ Plane Motion

- Position, Velocity and Acceleration vectors

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$$

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$$

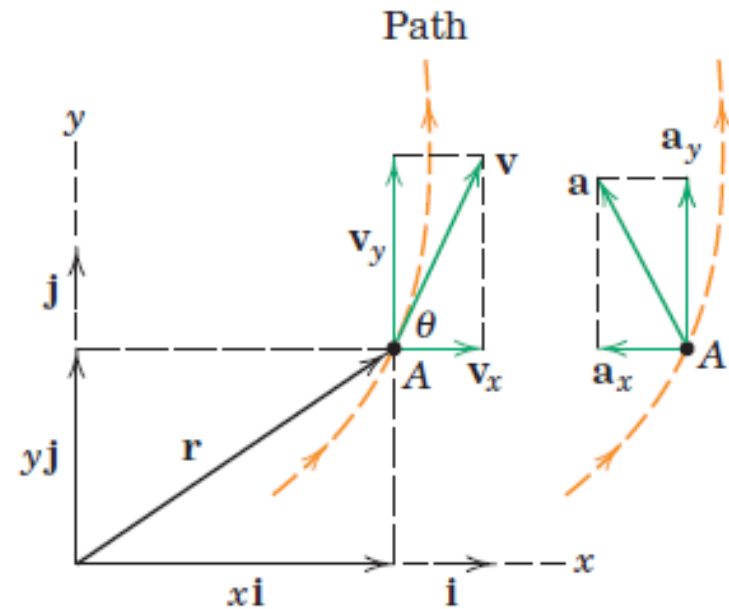
$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{\mathbf{r}} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}$$

$$v_x = \dot{x} \quad a_x = \dot{v}_x$$

$$v_y = \dot{y} \quad a_y = \dot{v}_y$$

$$v^2 = v_x^2 + v_y^2$$

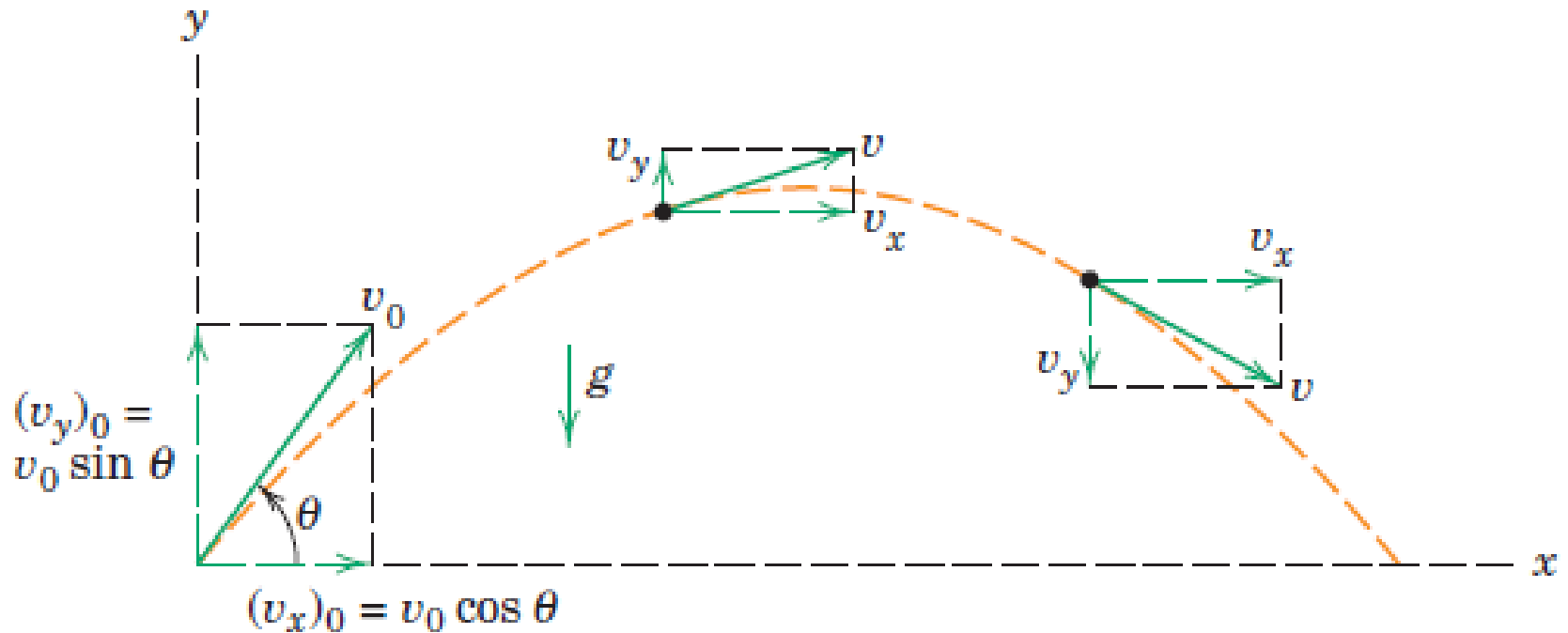
$$a^2 = a_x^2 + a_y^2$$



$$\tan \theta = \frac{v_y}{v_x} = \frac{dy}{dx}$$

Projectile Motion

■ Plane Motion



$$v_x = ? \quad a_x = ?$$

$$v_y = ? \quad a_y = ?$$

Projectile Motion

- **Plane Motion (study Curvilinear motion using rectilinear motion)**
- Position, Velocity and Acceleration vectors

$$v = v_0 + at \quad v^2 = v_0^2 + 2a(s - s_0) \quad s = s_0 + v_0 t + \frac{1}{2}at^2$$

$$a_x = 0$$

$$v_x = (v_x)_0$$

$$x = x_0 + (v_x)_0 t$$

$$a_y = -g$$

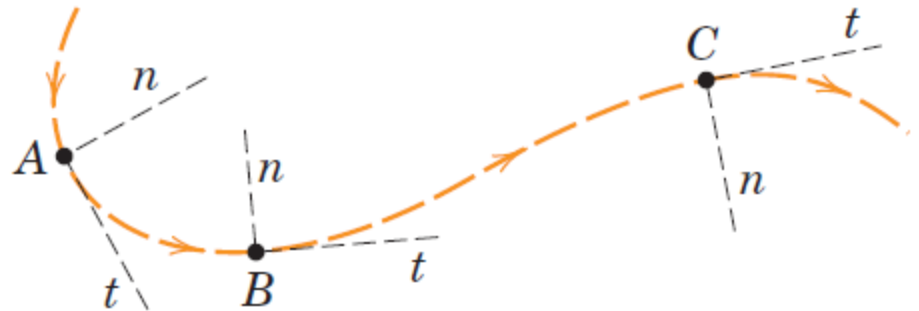
$$v_y = (v_y)_0 - gt$$

$$y = y_0 + (v_y)_0 t - \frac{1}{2}gt^2$$

$$v_y^2 = (v_y)_0^2 - 2g(y - y_0)$$

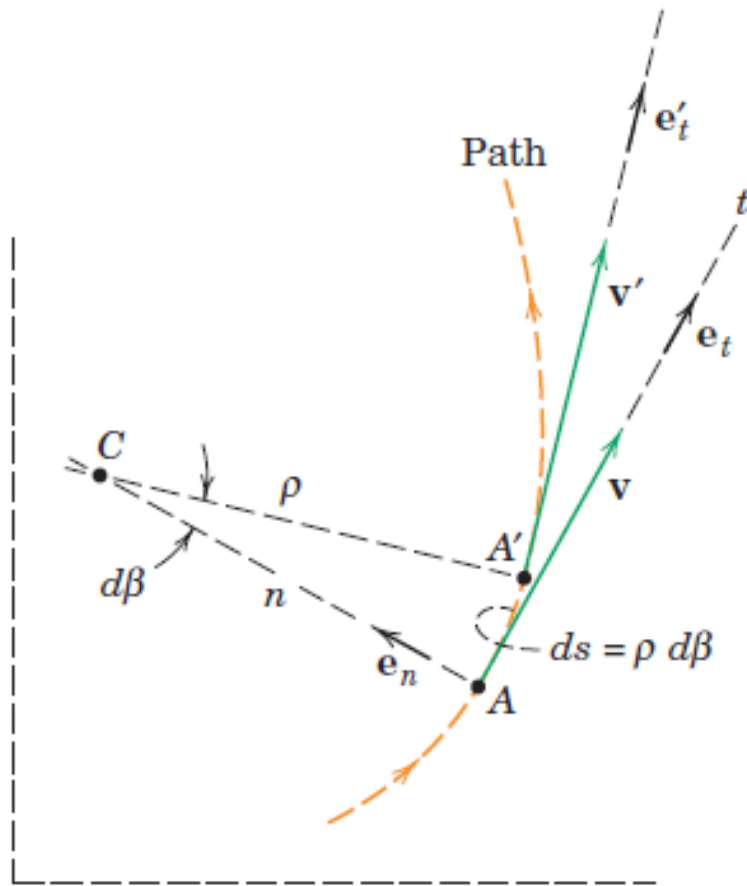
Normal and Tangential Coordinates (n-t)

- Path Variables:
Measurements along
normal and tangential
directions

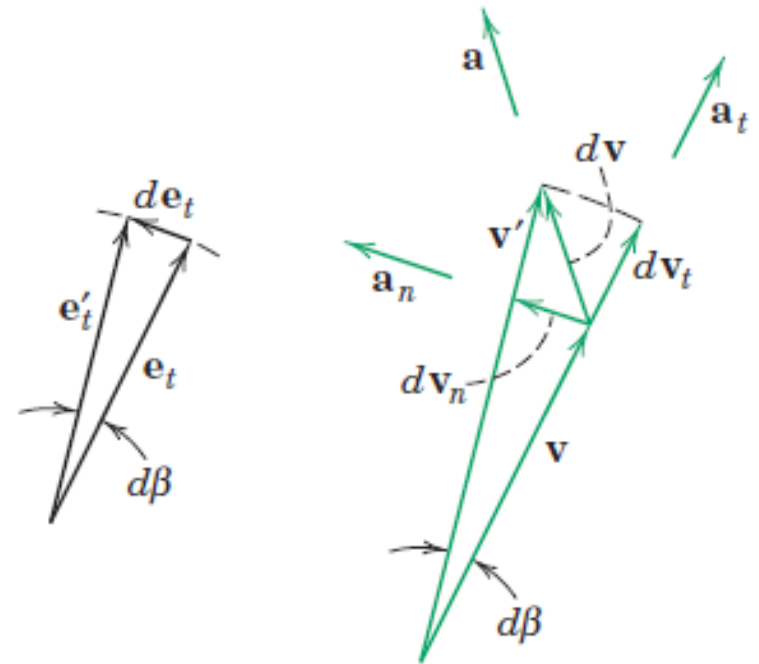


- Positive direction for n at any position is always taken toward the center of curvature of the path
- Positive n - direction will shift from one side of the curve to the other side if the curvature changes direction.

Normal and Tangential Coordinates (n-t)



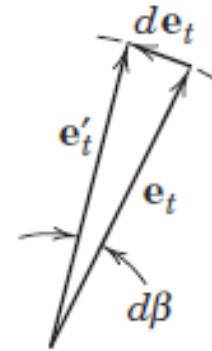
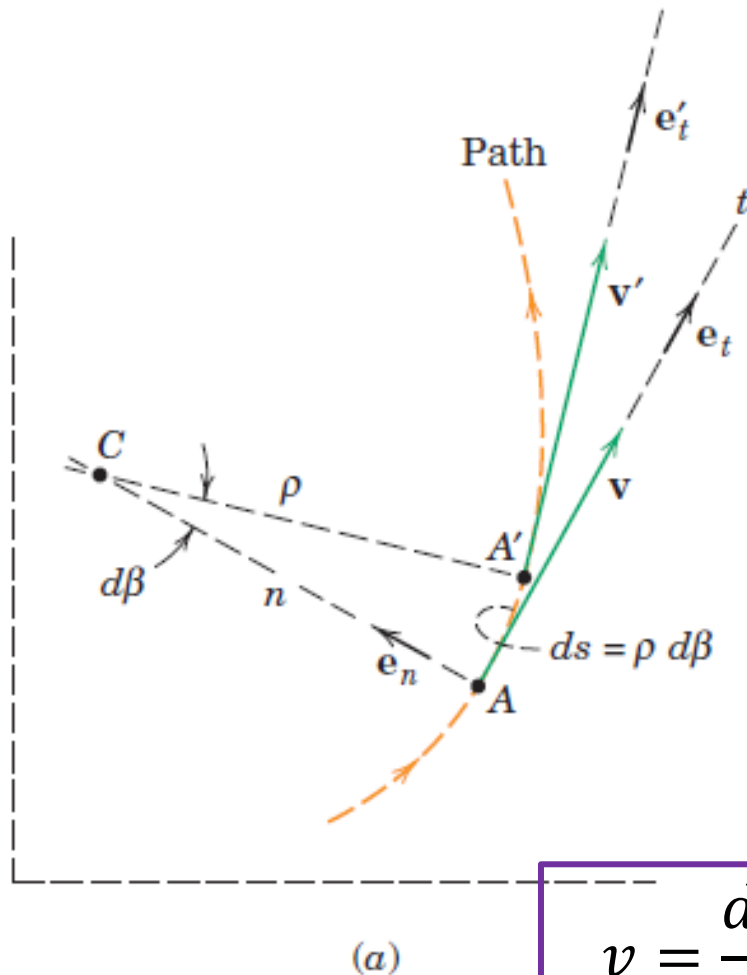
(a)



(b)

(c)

Normal and Tangential Coordinates (n-t)



$$|de_t| = d\beta \quad e_t \quad de_t = e_n d\beta$$

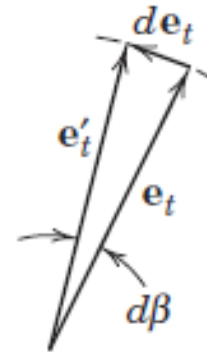
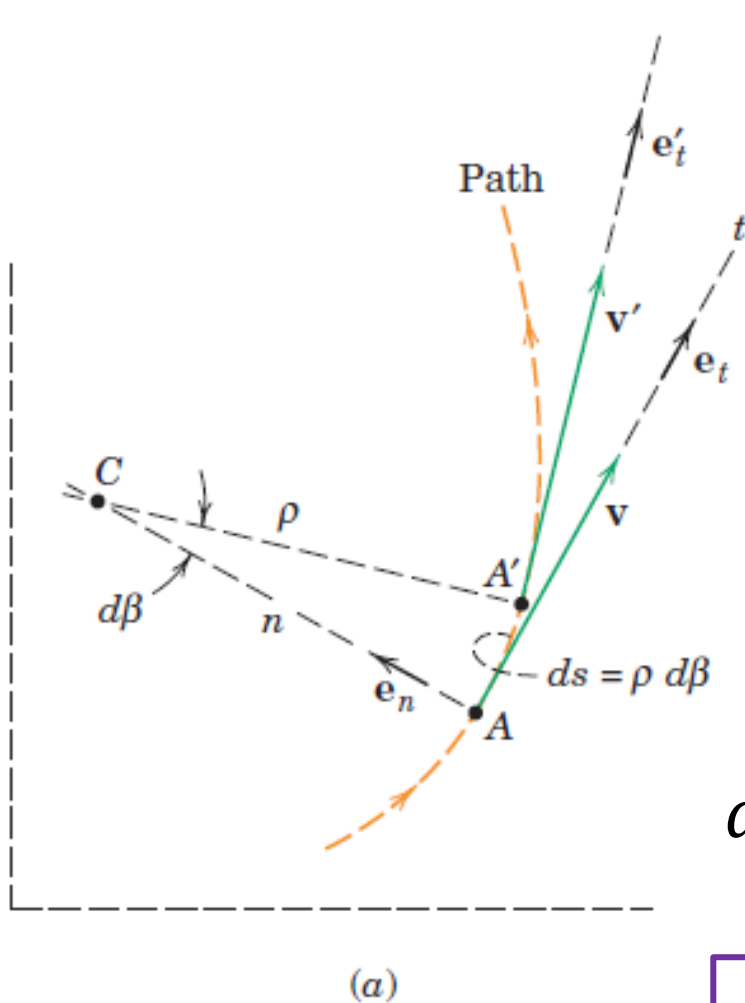
$$\frac{de_t}{dt} = e_n \frac{d\beta}{dt}$$

$$\dot{e}_t = e_n \dot{\beta}$$

$$v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho \dot{\beta}$$

$$\dot{e}_t = e_n \frac{v}{\rho}$$

Normal and Tangential Coordinates (n-t)



$$v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho \dot{\beta}$$

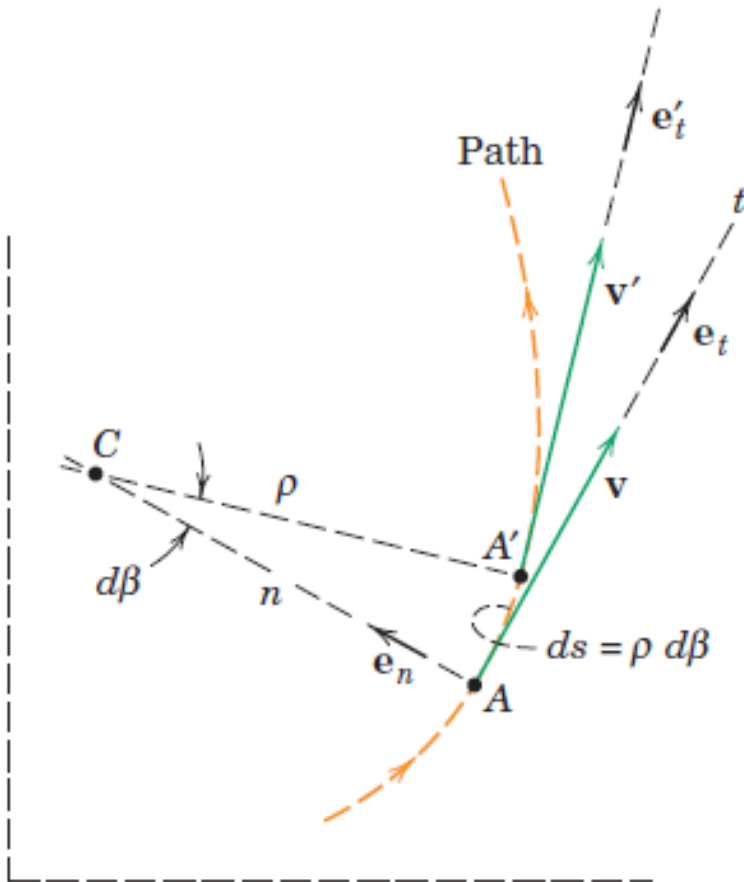
$$v = v e_t = \rho \dot{\beta} e_t$$

$$a = \frac{dv}{dt} = \frac{d(v e_t)}{dt} = \dot{v} e_t + v \dot{e}_t$$

$$\dot{e}_t = e_n \dot{\beta}$$

$$\dot{e}_t = e_n \frac{v}{\rho}$$

Normal and Tangential Coordinates (n-t)



$$v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho \dot{\beta}$$

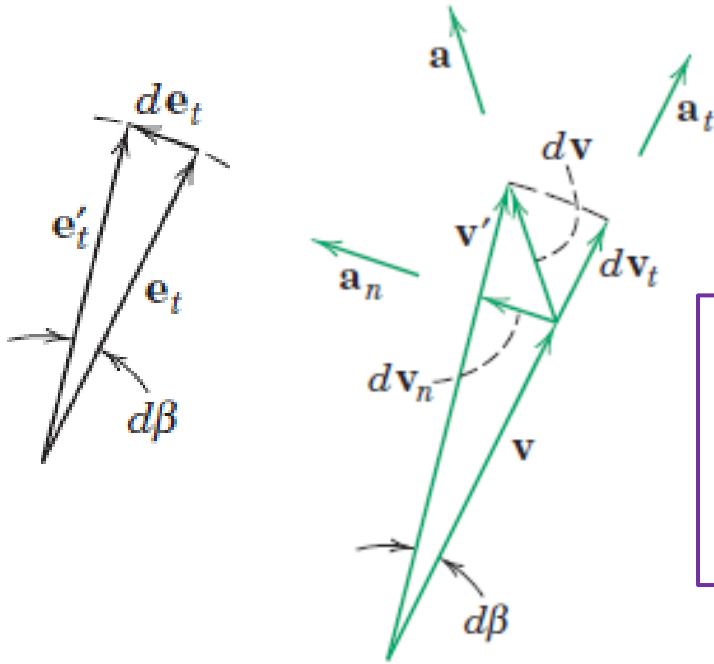
$$v = v e_t = \rho \dot{\beta} e_t$$

$$a = \frac{dv}{dt} = \frac{d(v e_t)}{dt} = \dot{v} e_t + v \dot{\beta} e_n$$

$$a_t = \dot{v} = \ddot{s}$$

$$a_n = v \dot{\beta} = \rho \dot{\beta}^2 = \frac{v^2}{\rho}$$

Normal and Tangential Coordinates (n-t)



$$v = \frac{ds}{dt} = \frac{\rho d\beta}{dt} = \rho \dot{\beta}$$

$$\mathbf{v} = v \mathbf{e}_t = \rho \dot{\beta} \mathbf{e}_t$$

$$d\mathbf{v}_n = |\mathbf{v}| d\beta$$

Acceleration contribution:

$$\frac{d\mathbf{v}_n}{dt} = \frac{|\mathbf{v}| d\beta}{dt} = |\mathbf{v}| \dot{\beta} = \rho \dot{\beta}^2 \text{ in } \mathbf{e}_n \text{ direction}$$

$$\mathbf{a} = \dot{v} \mathbf{e}_t + v \dot{\beta} \mathbf{e}_n$$

$d\mathbf{v}_t$: change in magnitude of $\mathbf{v}$

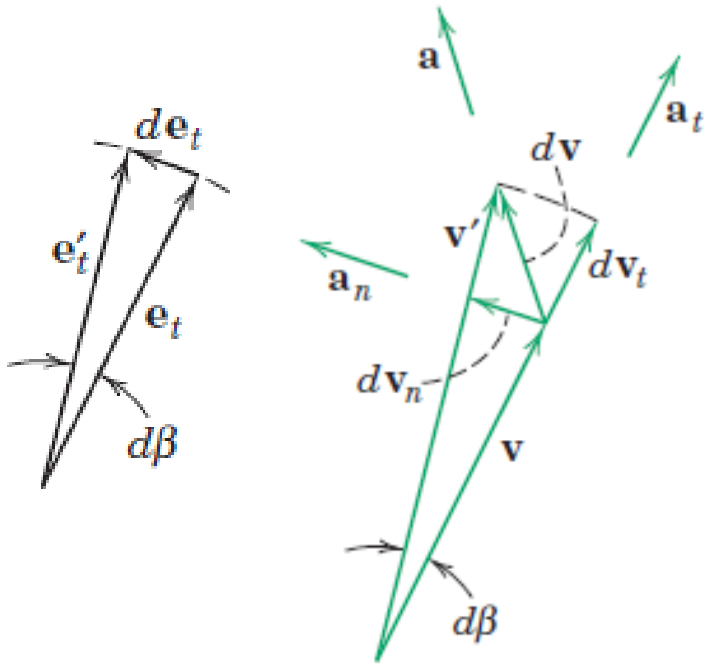
Acceleration contribution:

$$\frac{d\mathbf{v}_t}{dt} = \frac{d(\rho \dot{\beta})}{dt} \text{ in } \mathbf{e}_t \text{ direction}$$

$$\frac{d\mathbf{v}_t}{dt} = \ddot{s} \text{ in } \mathbf{e}_t \text{ direction}$$

Normal and Tangential Coordinates (n-t)

$$\mathbf{a} = \dot{v}\mathbf{e}_t + v\dot{\beta}\mathbf{e}_n$$



$$d\mathbf{v}_n = |\mathbf{v}|d\beta$$

Acceleration contribution:

$$\frac{d\mathbf{v}_n}{dt} = \frac{|\mathbf{v}|d\beta}{dt} = |\mathbf{v}|\dot{\beta} = \rho\dot{\beta}^2 \text{ in } \mathbf{e}_n \text{ direction}$$

$d\mathbf{v}_t$: change in magnitude of $\mathbf{v}$

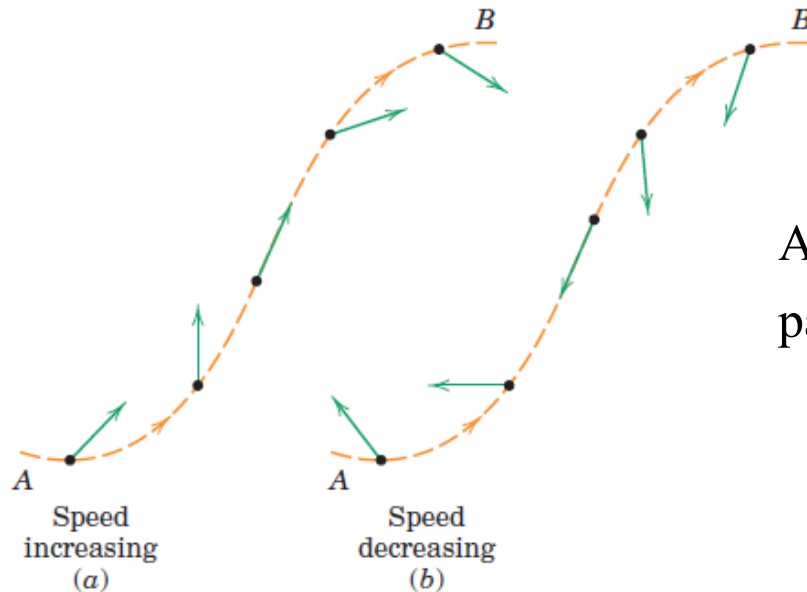
Acceleration contribution:

$$\frac{dv_t}{dt} = \frac{d(\rho\dot{\beta})}{dt} \text{ in } \mathbf{e}_t \text{ direction}$$

$$\frac{dv_t}{dt} = \ddot{s} \text{ in } \mathbf{e}_t \text{ direction}$$

- a_n is always directed toward the center of curvature C
- a_t positive t -direction of motion if the speed v is increasing and negative t -direction if the speed is decreasing.

Normal and Tangential Coordinates (n-t)



Acceleration vectors for
particle moving from A to B

(a) increasing speed and (b) decreasing speed

- For (a), tangential acceleration is in positive t -direction and for (b), tangential acceleration is in negative t -direction
- At an inflection point on the curve, the normal acceleration $\frac{v^2}{\rho}$ goes to zero because ρ becomes infinite

Normal and Tangential Coordinates (n-t)

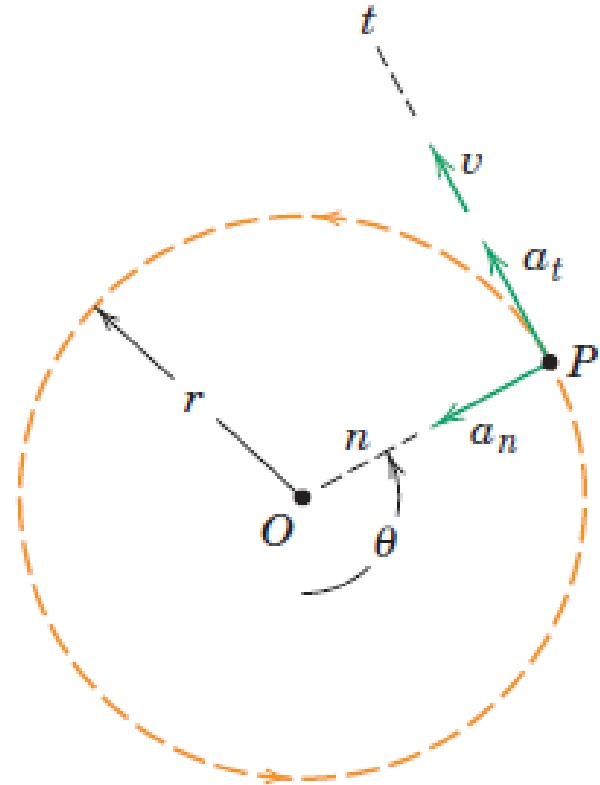
- Circular Motion**

$$v = r\dot{\theta} \quad \rho = r$$

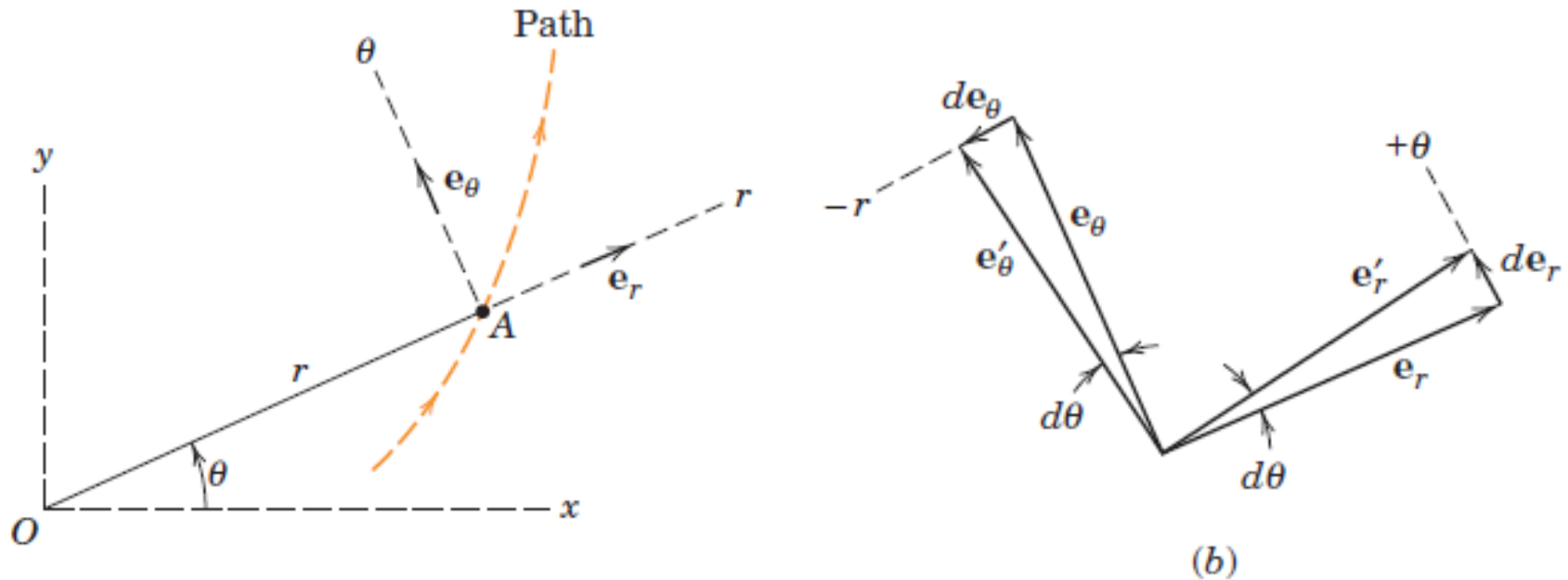
$$a = \dot{v}e_t + v\dot{\theta}e_n$$

$$a_t = \dot{v} = r\ddot{\theta}$$

$$a_n = \frac{v^2}{r} = r\dot{\theta}^2 = v\dot{\theta}$$



Polar Coordinates (r- θ)



$$\mathbf{r} = r\mathbf{e}_r \quad \frac{d\mathbf{e}_r}{d\theta} = \mathbf{e}_\theta \quad \frac{d\mathbf{e}_r}{dt} = \frac{d\theta}{dt}\mathbf{e}_\theta \quad \dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta$$

$$\frac{d\mathbf{e}_\theta}{d\theta} = -\mathbf{e}_r \quad \frac{d\mathbf{e}_\theta}{dt} = -\frac{d\theta}{dt}\mathbf{e}_r \quad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r$$

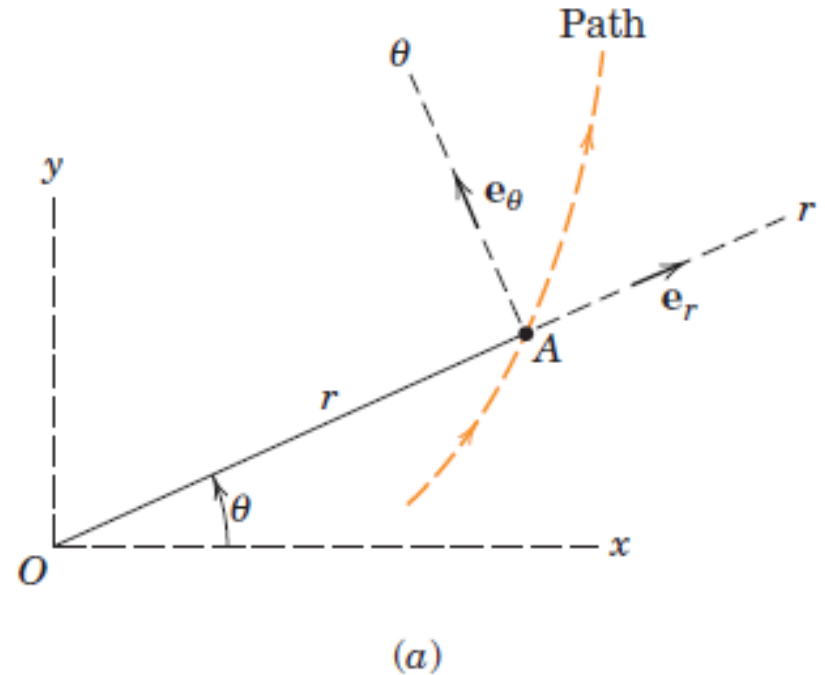
Polar Coordinates (r- θ)

$$\mathbf{r} = r\mathbf{e}_r$$

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r$$

$$\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta$$

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$$



- r - component of $\mathbf{v}$: rate at which the vector $\mathbf{r}$ stretches
- θ - component of $\mathbf{v}$: due to the rotation of $\mathbf{r}$

Polar Coordinates (r- θ)

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta \quad \dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta \quad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r$$

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta$$

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta - r\dot{\theta}\dot{\theta}\mathbf{e}_r$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

Polar Coordinates (r- θ)

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta$$

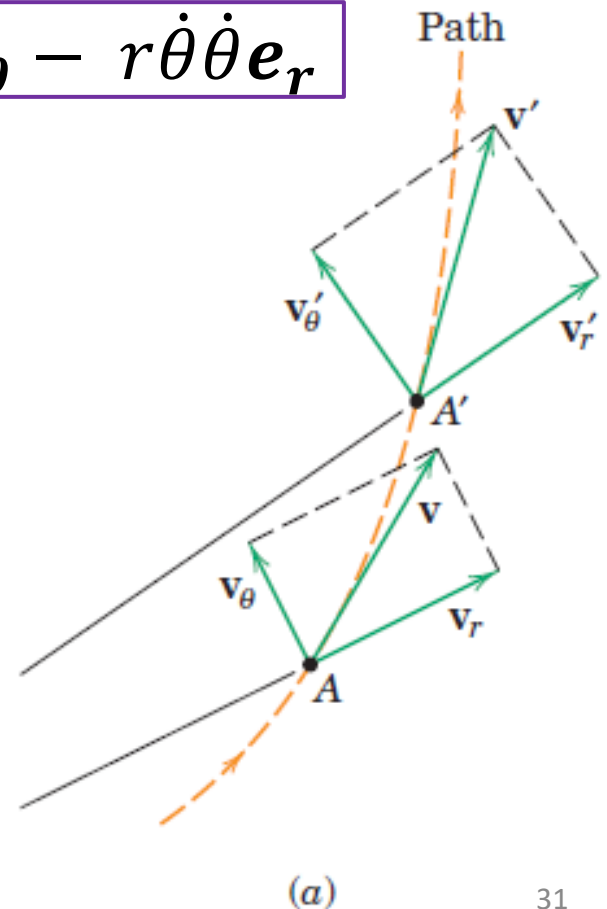
$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta - r\dot{\theta}\dot{\theta}\mathbf{e}_r$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

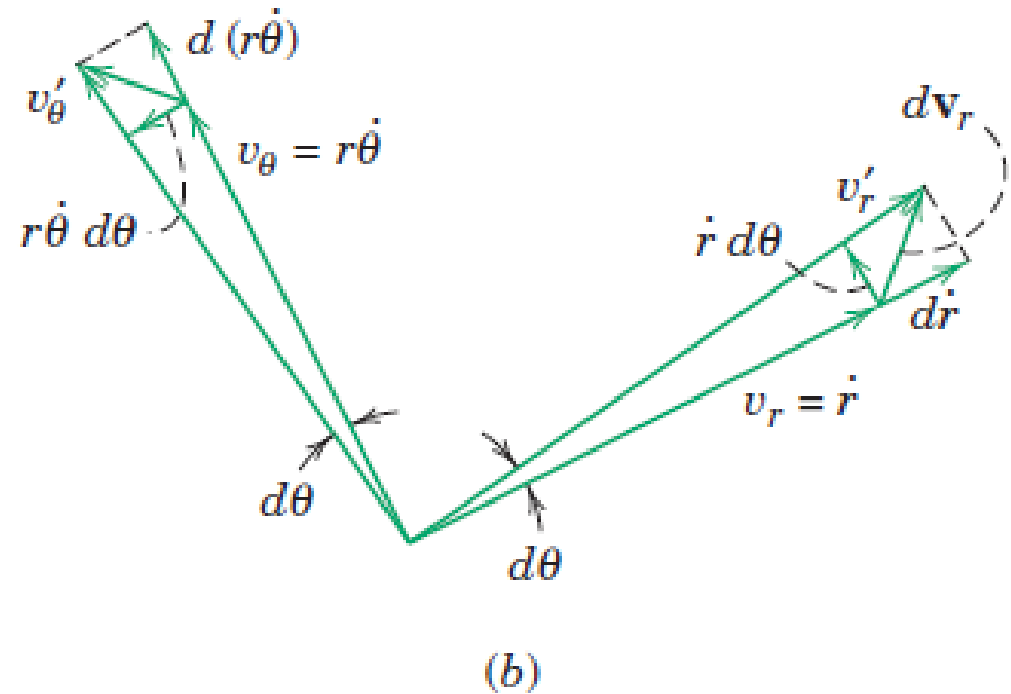
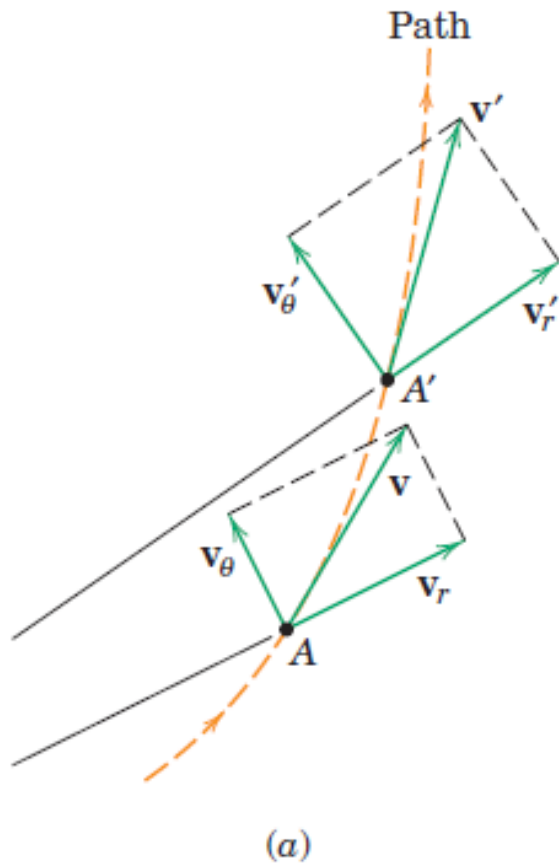
$$a_r = (\ddot{r} - r\dot{\theta}^2)$$

$$a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

$$a_\theta = \frac{1}{r} \frac{d(r^2\dot{\theta})}{dt}$$



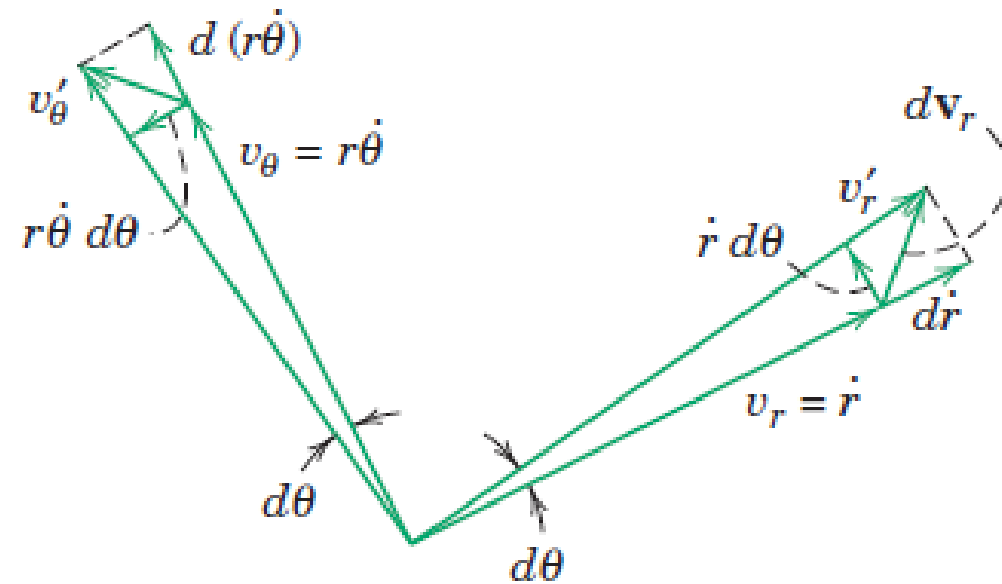
Polar Coordinates (r - θ)



Polar Coordinates (r- θ)

$$\mathbf{a} = \dot{\mathbf{v}} = (\ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r) + (\dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta)$$

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta - r\dot{\theta}\dot{\theta}\mathbf{e}_r$$



(b)

Magnitude change of v_r : $d\dot{r}$

Acceleration contribution:

$$\frac{d\dot{r}}{dt} = \ddot{r} \text{ in } \mathbf{e}_r \text{ direction}$$

Direction change of v_r : $\dot{r}d\theta$

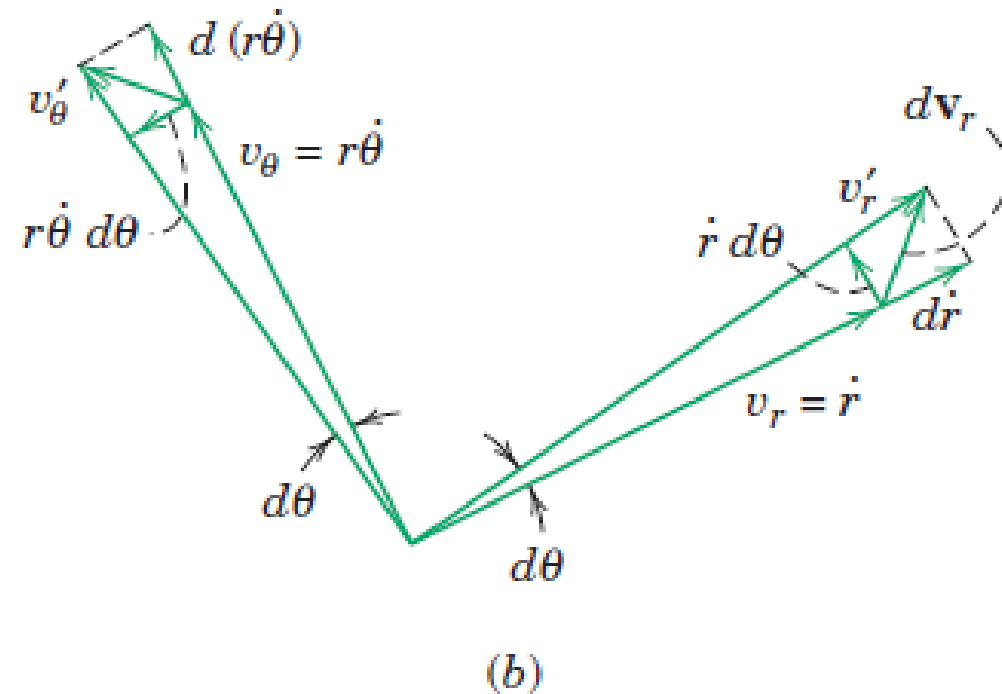
Acceleration contribution:

$$\frac{\dot{r}d\theta}{dt} = \dot{r}\dot{\theta} \text{ in } \mathbf{e}_\theta \text{ direction}$$

Polar Coordinates (r- θ)

$$\mathbf{a} = \dot{\mathbf{v}} = (\ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r) + (\dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta)$$

$$\mathbf{a} = \dot{\mathbf{v}} = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta - r\dot{\theta}\dot{\theta}\mathbf{e}_r$$



Magnitude change of v_θ : $d(r\dot{\theta})$

Acceleration contribution:

$$\frac{d(r\dot{\theta})}{dt} = r\ddot{\theta} + \dot{r}\dot{\theta} \text{ in } \mathbf{e}_\theta \text{ direction}$$

Direction change of v_θ : $r\dot{\theta}d\theta$

Acceleration contribution:

$$\frac{r\dot{\theta}d\theta}{dt} = r\dot{\theta}\ddot{\theta} \text{ in } -\mathbf{e}_r \text{ direction}$$

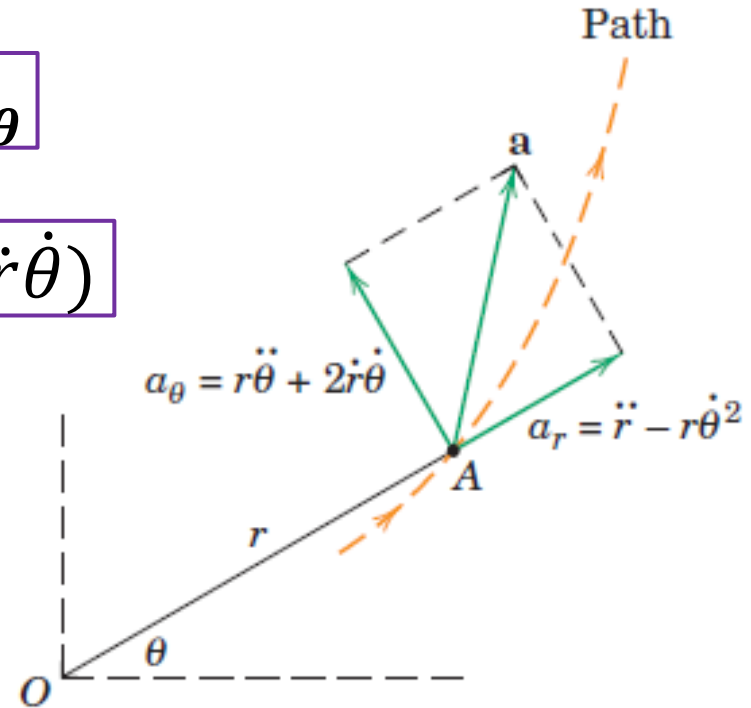
Polar Coordinates (r- θ)

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

$$a_r = (\ddot{r} - r\dot{\theta}^2)$$

$$a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

- $\ddot{r}$: acceleration in r-direction if θ is constant
- $-r\dot{\theta}^2$: acceleration in r-direction if r is constant as in circular motion
- $r\ddot{\theta}$: acceleration in θ -direction if r is constant as in circular motion
- $2\dot{r}\dot{\theta}$: acceleration in θ -direction due to change in direction of $\mathbf{v}_r$ (if r is not constant) and also due to change in magnitude of $\mathbf{v}_\theta$



Polar Coordinates (r- θ)

- Circular Motion

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$$

$$v_r = \dot{r} = 0$$

$$v_\theta = r\dot{\theta}$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

$$\ddot{r} = 0 \quad \dot{r} = 0$$

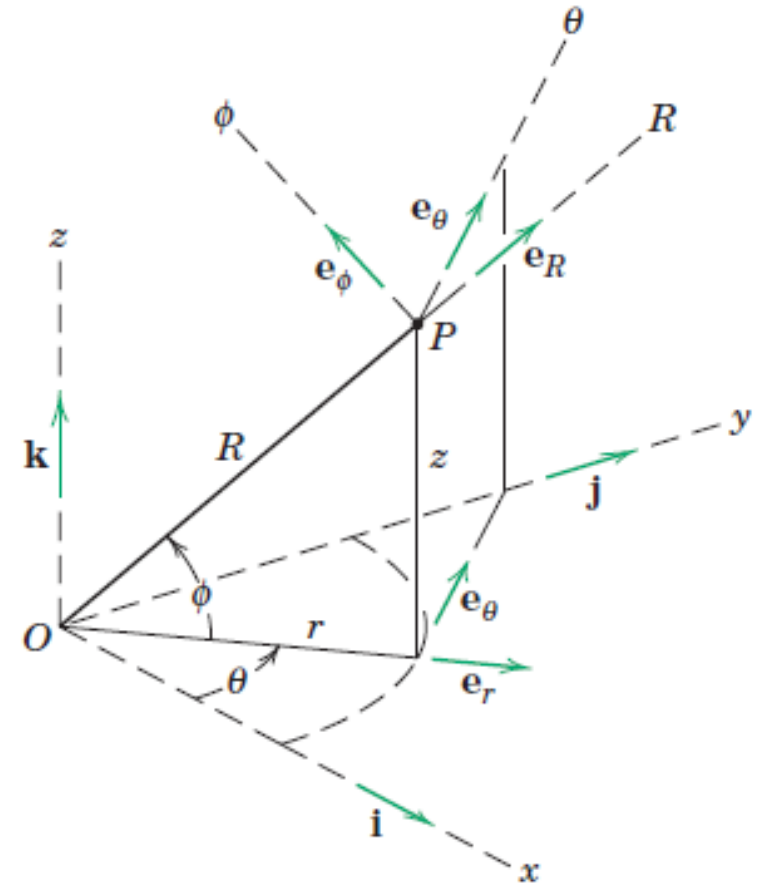
$$\mathbf{a} = (-r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta})\mathbf{e}_\theta$$

$$a_\theta = r\ddot{\theta}$$

$$a_r = -r\dot{\theta}^2$$

Space Curvilinear Motion

- Rectangular Coordinates (x-y-z)
- Cylindrical Coordinates (r- θ -z)
- Spherical Coordinates (r- θ - ϕ)



Space Curvilinear Motion

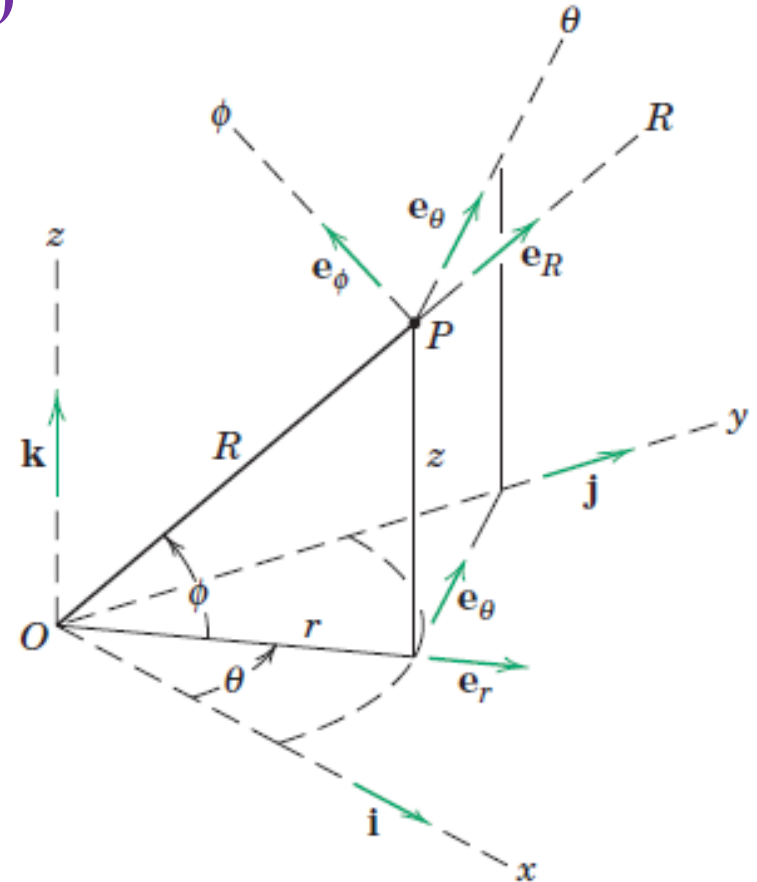
- Rectangular Coordinates (x-y-z)

$$R = xi + yj + zk$$

$$v = \dot{R} = \dot{x}i + \dot{y}j + \dot{z}k$$

$$a = \dot{v} = \ddot{R} = \ddot{x}i + \ddot{y}j + \ddot{z}k$$

R for position vector in place of r



Space Curvilinear Motion

■ Cylindrical Coordinates (r- θ -z)

$$\mathbf{R} = r\mathbf{e}_r + z\mathbf{k}$$

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{k}$$

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta + \ddot{z}\mathbf{k}$$

$$v_r = \dot{r}$$

$$v_\theta = r\dot{\theta}$$

$$v_z = \dot{z}$$

$$v^2 = v_r^2 + v_\theta^2 + v_z^2$$

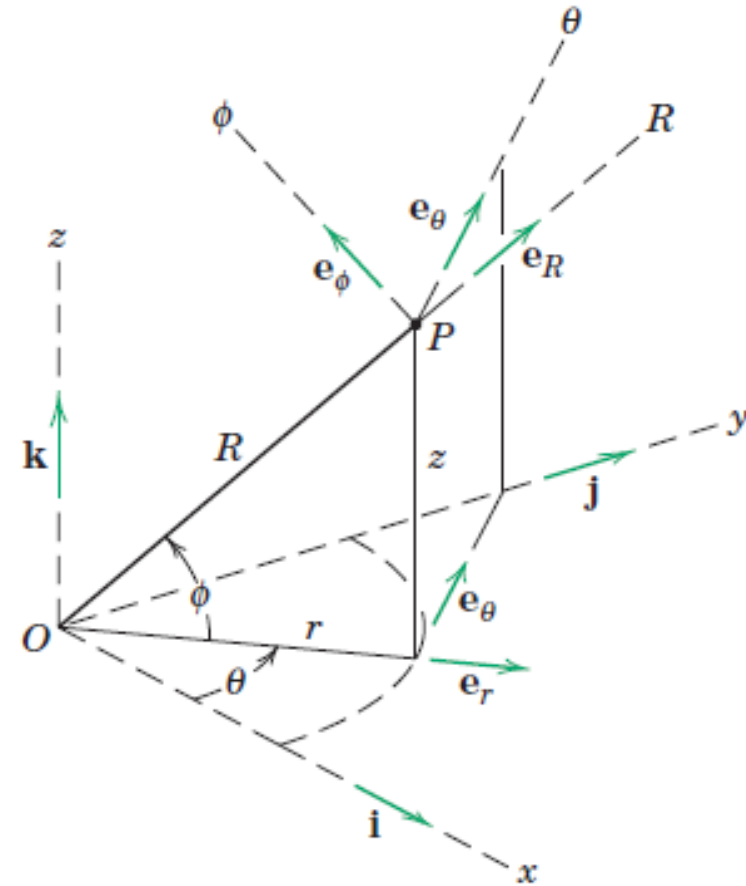
$$a_r = (\ddot{r} - r\dot{\theta}^2)$$

$$a_\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

$$a_z = \ddot{z}$$

$$a^2 = a_r^2 + a_\theta^2 + a_z^2$$

$\mathbf{e}_r$ and $\mathbf{e}_\theta$ have nonzero time derivatives, $\mathbf{e}_z$ has zero time derivative



Space Curvilinear Motion

■ Spherical Coordinates (R- θ - ϕ)

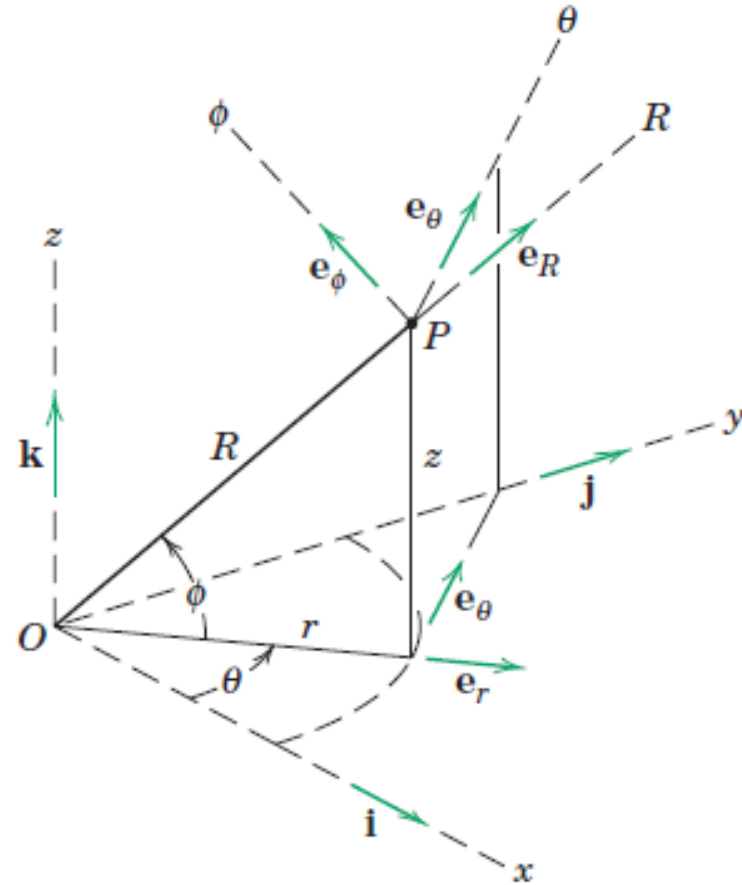
$$\mathbf{v} = v_R \mathbf{e}_R + v_\theta \mathbf{e}_\theta + v_\phi \mathbf{e}_\phi$$

$$v_R = \dot{R}$$

$$v_\theta = R \dot{\theta} \cos \phi$$

$$v_\phi = R \dot{\phi}$$

$$v^2 = v_R^2 + v_\theta^2 + v_\phi^2$$



Space Curvilinear Motion

■ Spherical Coordinates (R- θ - ϕ)

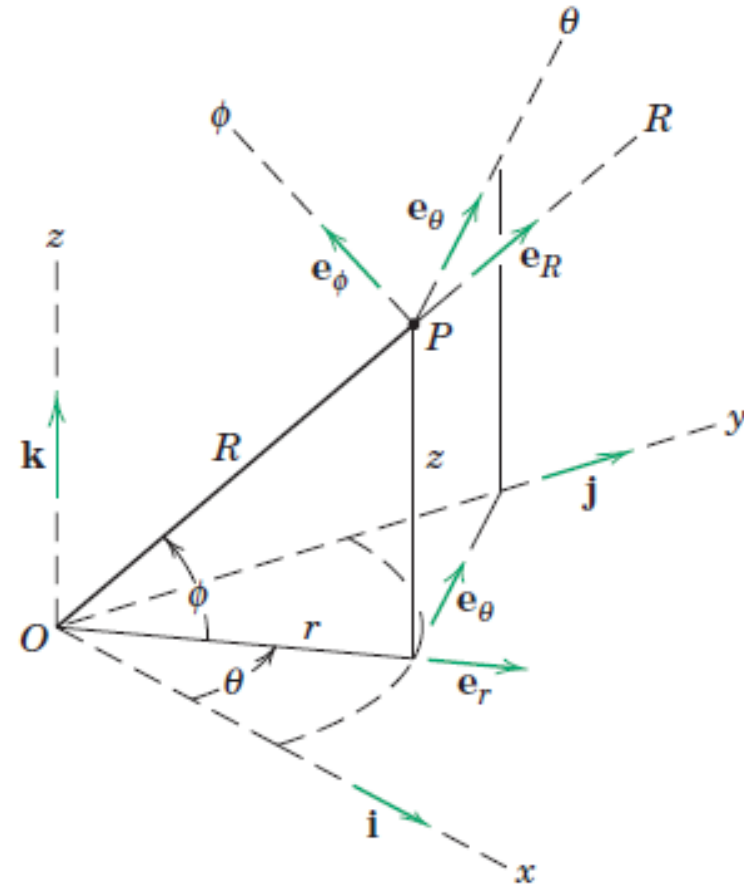
$$\mathbf{a} = a_R \mathbf{e}_R + a_\theta \mathbf{e}_\theta + a_\phi \mathbf{e}_\phi$$

$$a_R = \ddot{R} - R\dot{\phi}^2 - R\dot{\theta}^2 (\cos \phi)^2$$

$$a_\theta = \frac{\cos \phi}{R} \frac{d(R^2 \dot{\theta})}{dt} - 2R\dot{\theta}\dot{\phi} \sin \phi$$

$$a_\phi = \frac{1}{R} \frac{d(R^2 \dot{\phi})}{dt} + R\dot{\theta}^2 \sin \phi \cos \phi$$

$$a^2 = a_r^2 + a_\theta^2 + a_\phi^2$$



Space Curvilinear Motion

- Which coordinate system should be used?
 - Rectangular Coordinates
(x - y - z)
 - Cylindrical Coordinates
(r - θ - z)
 - Spherical Coordinates
(r - θ - ϕ)



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A portion of the track of this amusement-park ride is in the shape of a helix whose axis is horizontal.

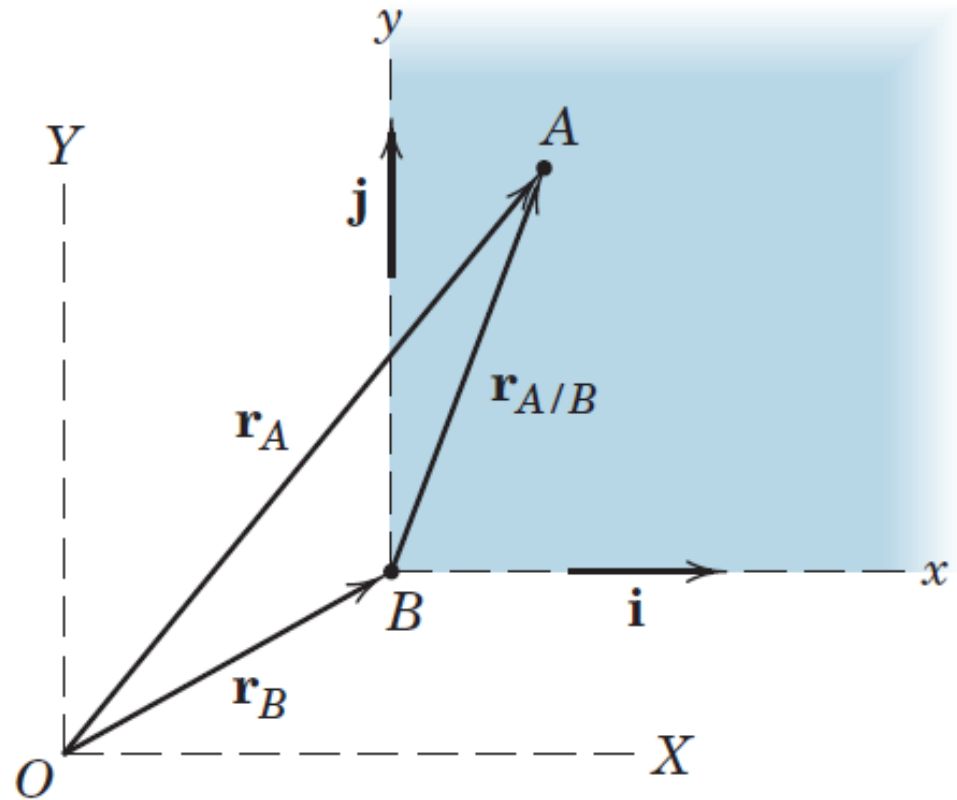
Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

Position vector of A as measured relative to the frame x - y

$$\mathbf{r}_{A/B} = x\mathbf{i} + y\mathbf{j}$$

“ A/B ” means “ A relative to B ”
or “ A with respect to B ”



Relative Motion (Translating Axis)

Moving reference system translate but do not rotate

$$\mathbf{r}_{A/B} = x\mathbf{i} + y\mathbf{j}$$

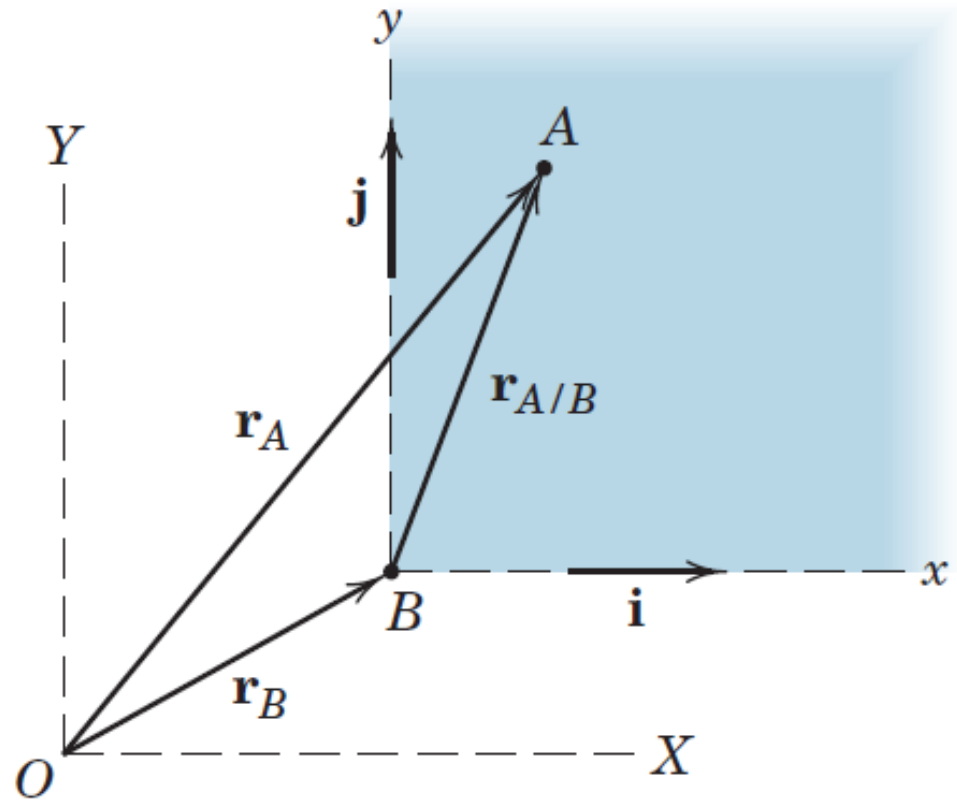
$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}_{A/B}$$

$$\dot{\mathbf{r}}_A = \dot{\mathbf{r}}_B + \dot{\mathbf{r}}_{A/B}$$

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$

$$\ddot{\mathbf{r}}_A = \ddot{\mathbf{r}}_B + \ddot{\mathbf{r}}_{A/B}$$

$$\mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B}$$



THANK YOU